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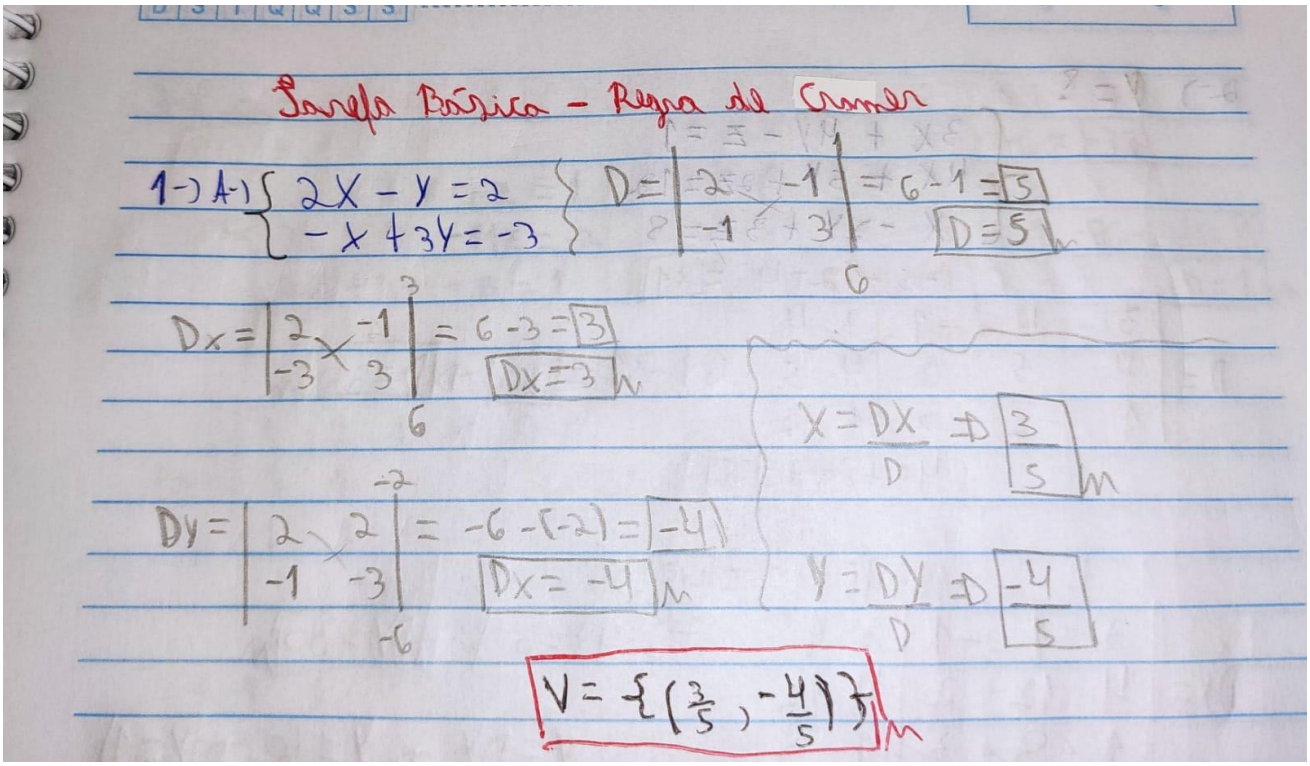
Sistemas Lineares - Regra de Cramer - Escalonamento (Gauss)

(Fotos nas páginas seguintes)

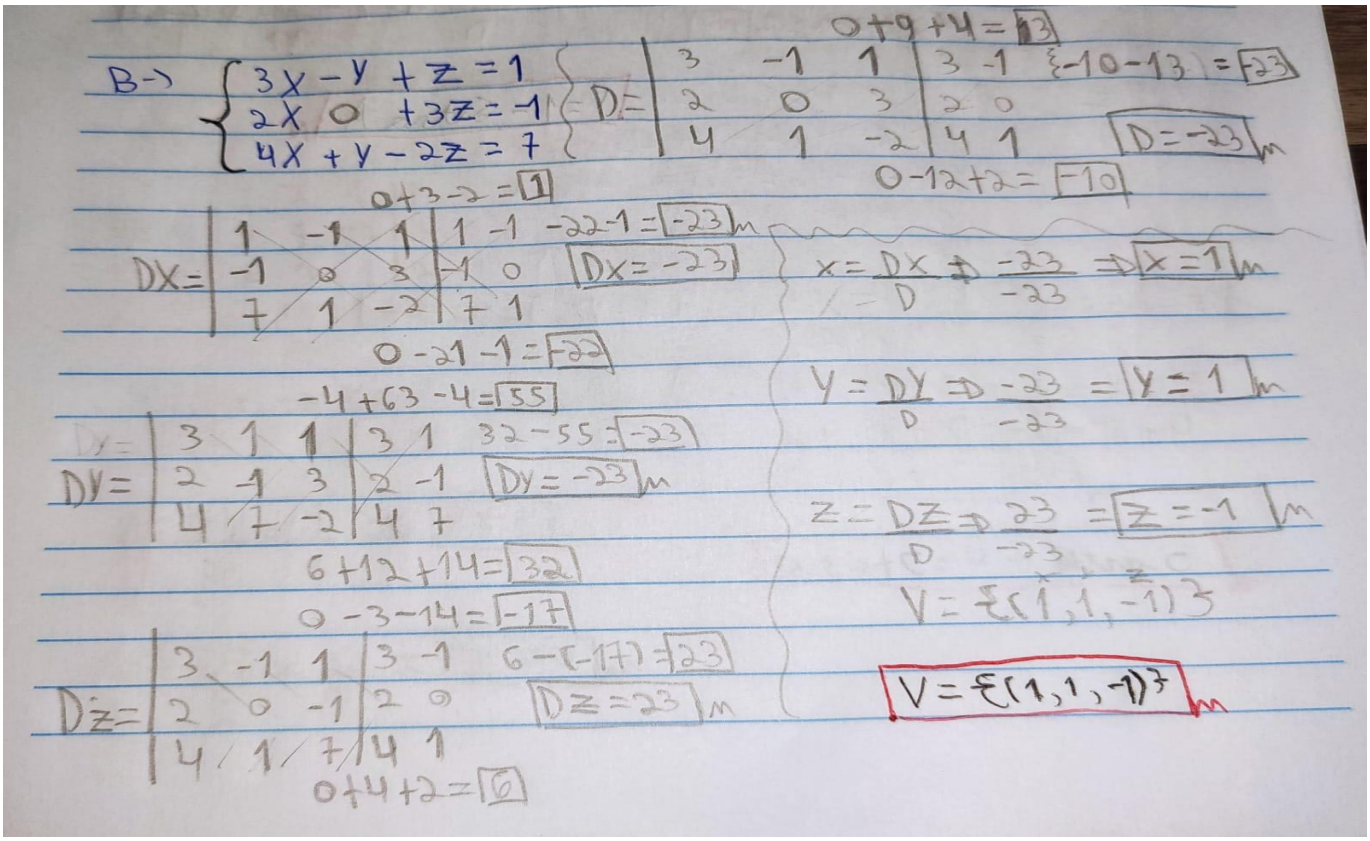
Regra de Cramer

Exercício 1 – A e B:

A-)



B-)



Exercício 2:

2-) $Y = ?$

$$\begin{cases} 3X + 4Y - Z = 1 \\ 4X + 5Y + 2Z = 12 \\ X - 2Y + 3Z = 8 \end{cases}$$

$-5 - 12 + 48 = \boxed{31}$

$$D = \left|\begin{matrix} 3 & 4 & -1 \\ 4 & 5 & 2 \\ 1 & -2 & 3 \end{matrix}\right| \begin{matrix} 3 & 4 \\ 4 & 5 \\ 1 & -2 \end{matrix}$$

$61 - 31 = \boxed{30} \Rightarrow \boxed{D = 30}$

$45 + 8 + 8 = \boxed{61}$

$-12 + 48 + 12 = \boxed{48}$

$$DY = \left|\begin{matrix} 3 & 1 & -1 \\ 4 & 12 & 2 \\ 1 & 8 & 3 \end{matrix}\right| \begin{matrix} 3 & 1 \\ 4 & 12 \\ 1 & 8 \end{matrix}$$

$48 - 48 = \boxed{30} \Rightarrow \boxed{DY = 30}$

$Y = \frac{DY}{D} \Rightarrow Y = \frac{30}{30} \Rightarrow \boxed{Y = 1}$

$108 + 2 - 32 = \boxed{78}$

$\boxed{Y = 1 - \text{Letra A.}}$

Exercício 3:

3-) $a = X \mid b = Y \mid c = Z$

$$\begin{cases} x + 2y + z = 1 \\ 3x + y - 11z = -2 \\ 2x + 3y - z = 1 \end{cases}$$

$$D = \left|\begin{array}{ccc|cc} 1 & 2 & 1 & 1 & 2 \\ 3 & 1 & -11 & 3 & 1 \\ 2 & 3 & -1 & 2 & 3 \end{array}\right.$$

$2 - 33 - 6 = \boxed{-37}$

$-1 - 44 + 9 = \boxed{-36}$

$-36 - (-37) = \boxed{1}$

$\boxed{D = 1}$

$1 - 33 + 4 = \boxed{-28}$

$$Dx = \left|\begin{array}{ccc|cc} 1 & 2 & 1 & 1 & 2 \\ -2 & 1 & -11 & -2 & 1 \\ 1 & 3 & -1 & 1 & 3 \end{array}\right.$$

$-29 - (-28) = -1$

$\boxed{Dx = -1}$

$-1 - 22 - 6 = \boxed{-29}$

$-4 - 11 - 3 = \boxed{-18}$

$$Dy = \left|\begin{array}{ccc|cc} 1 & 1 & 1 & 1 & 1 \\ 3 & -2 & -11 & 3 & -2 \\ 2 & 1 & -1 & 2 & 1 \end{array}\right.$$

$-11 - (-18) = \boxed{1}$

$\boxed{Dy = 1}$

$2 - 22 + 3 = \boxed{-17}$

$2 - 6 + 6 = \boxed{2}$

$$Dz = \left|\begin{array}{ccc|cc} 1 & 2 & 1 & 1 & 2 \\ 3 & 1 & -2 & 3 & 1 \\ 2 & 3 & 1 & 2 & 3 \end{array}\right.$$

$2 - 2 = \boxed{0}$

$\boxed{Dz = 0}$

$1 - 8 + 9 = \boxed{2}$

$X = \frac{Dx}{D} \Rightarrow X = \frac{-1}{1} \Rightarrow \boxed{X = -1}$

$Y = \frac{Dy}{D} = \frac{1}{1} \Rightarrow \boxed{Y = 1}$

$Z = \frac{Dz}{D} = \frac{0}{1} = \boxed{Z = 0}$

$a + b + c = X + Y + Z$

$a + b + c = -1 + 1 + 0$

$\boxed{a + b + c = 0}$

$\boxed{a + b + c = 0 \text{ - letra c.}}$

Exercício 4:

4-) $(x+y+z)=?$

$$\begin{cases} x+2y-3z=29 \\ x+3y+2z=4 \\ x-y-2z=8 \end{cases}$$

$$D=\begin{vmatrix} 1 & 2 & -3 \\ 1 & 3 & 2 \\ 1 & -1 & -2 \end{vmatrix}\begin{matrix} 1 & 2 \\ 1 & 3 \\ 1 & -1 \end{matrix}$$

$-9-2-4=\boxed{-15}$

$-6+4+3=\boxed{1}$

$1-(-15)=\boxed{16}$

$\boxed{D=16}$

$-12-58-16=\boxed{-146}$

$$Dx=\begin{vmatrix} 29 & 2 & -3 \\ 4 & 3 & 2 \\ 8 & -1 & -2 \end{vmatrix}\begin{matrix} 29 & 2 \\ 4 & 3 \\ 8 & -1 \end{matrix}$$

$-174+32+12=\boxed{-130}$

$-130-(-146)=\boxed{16}$

$\boxed{Dx=16}$

$-12+16-58=\boxed{-54}$

$$Dy=\begin{vmatrix} 1 & 29 & -3 \\ 1 & 4 & 2 \\ 1 & 8 & -2 \end{vmatrix}\begin{matrix} 1 & 29 \\ 1 & 4 \\ 1 & 8 \end{matrix}$$

$-8+58-24=\boxed{26}$

$26-(-54)=\boxed{80}$

$\boxed{Dy=80}$

$87-4+16=\boxed{99}$

$$Dz\begin{vmatrix} 1 & 2 & 29 \\ 1 & 3 & 4 \\ 1 & -1 & 8 \end{vmatrix}\begin{matrix} 1 & 2 \\ 1 & 3 \\ 1 & -1 \end{matrix}$$

$24+8-29=\boxed{3}$

$3-99=\boxed{-96}$

$\boxed{Dz=-96}$

$x=\frac{16}{16}=\boxed{1}$ $\quad y=\frac{80}{16}=\boxed{5}$ $\quad z=\frac{-96}{16}=\boxed{-6}$

$x+y+z=?\rightarrow\boxed{x+y+z=0}$

$1+5-6=?$

$\boxed{6-6=0}$

$\boxed{x+y+z=0 - \text{letra A}}$

$\boxed{\text{letra} - 0=7+0+0}$

Exercício 5:

5 -> $X = ?$ | $Y = ?$ | $Z = ?$

$$\begin{cases} 2X + Y + 0 = 5 \\ 0 + 2Y + Z = 3 \\ 3X + 2Y + Z = 7 \end{cases}$$

$$D = \begin{vmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 3 & 2 & 1 \end{vmatrix} \begin{matrix} 2 & 1 \\ 0 & 2 \\ 3 & 2 \end{matrix}$$

$0 + 4 + 0 = \boxed{4}$

$4 + 3 + 0 = \boxed{7}$

$7 - 4 = 3$

$\boxed{D = 3}$

$0 + 10 + 3 = \boxed{13}$

$$Dx = \begin{vmatrix} 5 & 1 & 0 \\ 3 & 2 & 1 \\ 7 & 2 & 1 \end{vmatrix} \begin{matrix} 5 & 1 \\ 3 & 2 \\ 7 & 2 \end{matrix}$$

$17 - 13 = \boxed{4}$

$\boxed{Dx = 4}$

$10 + 7 + 0 = \boxed{17}$

$0 + 14 + 0 = \boxed{14}$

$$Dy = \begin{vmatrix} 2 & 5 & 0 \\ 0 & 3 & 1 \\ 3 & 7 & 1 \end{vmatrix} \begin{matrix} 2 & 5 \\ 0 & 3 \\ 3 & 7 \end{matrix}$$

$21 - 14 = \boxed{7}$

$\boxed{Dy = 7}$

$6 + 15 + 0 = \boxed{21}$

$30 + 12 + 0 = \boxed{42}$

$$Dz = \begin{vmatrix} 2 & 1 & 5 \\ 0 & 2 & 3 \\ 3 & 2 & 7 \end{vmatrix} \begin{matrix} 2 & 1 \\ 0 & 2 \\ 3 & 2 \end{matrix}$$

$37 - 42 = \boxed{-5}$

$\boxed{Dz = -5}$

$28 + 9 + 0 = \boxed{37}$

$X = \frac{Dx}{D} = \frac{4}{3} \Rightarrow \boxed{X = \frac{4}{3}}$

$\boxed{X = \frac{4}{3}}$

$Y = \frac{Dy}{D} = \frac{7}{3} \Rightarrow \boxed{Y = \frac{7}{3}}$

$\boxed{Y = \frac{7}{3}}$

$\boxed{\text{Letra D}}$

$Z = \frac{Dz}{D} \Rightarrow \boxed{Z = \frac{5}{3}}$

$\boxed{Z = \frac{5}{3}}$

$V = \{(\frac{4}{3}; \frac{7}{3}; \frac{5}{3})\}$

Exercício 6:

6-) $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 2 \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 3 \\ 7 \\ -1 \end{bmatrix}$

$\begin{vmatrix} x \\ y \\ z \end{vmatrix}$

$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} \begin{matrix} 1.x & y.0 & 0.z \\ 2.x & 1.y & 0.z \\ -1x & 2.y & 2.z \end{matrix}$

$\Rightarrow \begin{cases} x \quad 0 \quad 0 = 3 \\ 2x \quad y \quad 0 = 7 \\ -x \quad 2y \quad 2z = 1 \end{cases}$

$0+0+0=\boxed{0}$

$D = \begin{vmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} \begin{matrix} 1 & 0 \\ 2 & 1 \\ -1 & 2 \end{matrix}$ $2-0=\boxed{2}$ $\boxed{D=2}$

$2+0+0=\boxed{2}$

$0+0+0=\boxed{0}$

$Dx = \begin{vmatrix} 3 & 0 & 0 \\ 7 & 1 & 0 \\ -1 & 2 & 2 \end{vmatrix} \begin{matrix} 3 & 0 \\ 7 & 1 \\ -1 & 2 \end{matrix}$ $6-0=\boxed{6}$ $\boxed{Dx=6}$

$6+0+0=\boxed{6}$

$0+0+0=\boxed{0}$

$Dy = \begin{vmatrix} 1 & 3 & 0 \\ 2 & 7 & 0 \\ -1 & -1 & 2 \end{vmatrix} \begin{matrix} 1 & 3 \\ 2 & 7 \\ -1 & -1 \end{matrix}$ $14-0=\boxed{14}$ $\boxed{Dy=14}$

$14+0+0=\boxed{14}$

$-3+11+0=11$

$Dz = \begin{vmatrix} 1 & 0 & 3 \\ 2 & 1 & 7 \\ -1 & 2 & -1 \end{vmatrix} \begin{matrix} 1 & 0 \\ 2 & 1 \\ -1 & 2 \end{matrix}$ $11-11=0$ $\boxed{Dz=0}$

$-1+0+12=\boxed{11}$

$x = \frac{Dx}{D} = \frac{6}{2} \Rightarrow \boxed{x=3}$

$z = \frac{Dz}{D} = \frac{0}{2} = \boxed{z=0}$

$y = \frac{Dy}{D} = \frac{14}{2} \Rightarrow \boxed{y=7}$

$\boxed{z=0 \checkmark \text{ - Letra E}}$

Escalonamento (Gauss)

Exercício 1:

Lista Básica - Escalonamento (Gauss)

1-)

$$S \begin{cases} 2x - y - 3z = -5 \\ x + 3y - z = 11 \\ x + 0 - 5z = 3 \end{cases} \Rightarrow \left(\begin{array}{ccc|c} 2 & -1 & -3 & -5 \\ 1 & 3 & -1 & 11 \\ 1 & 0 & -5 & 3 \end{array}\right) \therefore$$

$$\therefore \begin{array}{c} \\ +1 \\ -1 \end{array} \left(\begin{array}{ccc|c} & & & \\ 1 & 0 & -10 & -4 \\ 1 & 0 & -5 & 3 \end{array}\right) \sim \left(\begin{array}{ccc|c} & & & \\ & & & \\ 0 & 0 & 25 & -25 \end{array}\right)$$

$$25z = -25 \Rightarrow \boxed{z = -1}$$

$$z = \frac{-25}{25}$$

$$1x - 10z = -4$$
$$1x - 10 \cdot -1 = -4$$
$$1x + 10 = -4$$
$$1x = -4 - 10$$
$$1x = -14$$
$$x = \frac{-14}{1} \Rightarrow \boxed{x = -2}$$

$$2x - y - 3z = -5$$
$$2 \cdot (-2) - y - 3 \cdot (-1) = -5$$
$$-4 - y + 3 = -5$$
$$-4 + 5 + 3 = y$$
$$\boxed{y = 4}$$

$$\boxed{x = -2 \; ; \; y = 4 \; ; \; z = -1}$$

Exercício 2:

2-) $\begin{cases} x = 2y \\ 2y = 3z \\ x + y + z = 11 \end{cases}$ $\Rightarrow 2y = 3z \Rightarrow \boxed{z = \frac{2y}{3}}$

$\boxed{x = 2y}$

$x + y + z = 11$

$2y + y + \frac{2y}{3} = 11 \quad (\cdot 3)$

$6y + 3y + 2y = 33$

$11y = 33$

$y = 33/11$

$\boxed{y = 3}$ → Substituir

$x = 2y$

$x = 2 \cdot 3$

$\boxed{x = 6}$

$2y = 3z$

$2 \cdot 3 = 3z$

$6 = 3z$

$\frac{6}{3} = z$

$\boxed{z = 2}$

$x + 2y + 3z = ?$ → $\boxed{x + 2y + 3z = 18}$

$6 + 2 \cdot 3 + 3 \cdot 2 = ?$

$6 + 6 + 6 = \boxed{18}$

Letra B.

Exercício 3:

3-) $z = ?$

$$\begin{cases} x + y + z = 0 \\ 2x - y - 2z = 1 \\ 6y + 3z \end{cases}$$

$$\begin{array}{l} -2\cdot \\ \\ \end{array}\left(\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 2 & -1 & -2 & 1 \\ 0 & 6 & 3 & -12 \end{array}\right) \therefore$$

$$\therefore \begin{array}{l} 2\cdot \\ 1\cdot \end{array}\left(\begin{array}{ccc|c} 0 & -3 & -4 & 1 \\ 0 & 6 & 3 & -12 \end{array}\right) \sim \left(\begin{array}{ccc|c} & & & \\ 0 & 0 & -5 & -10 \end{array}\right) \therefore$$

$$\therefore -5z = -10$$

$$z = \frac{-10}{-5}$$

$$\boxed{z = 2}$$

$$\boxed{z = 2 \text{ - Letra D}}$$

Exercício 4:

D S T Q Q S S

4 -> A = Cali | B = Bia | C = Coco. $\boxed{A + B + C = 68}$

$B + \frac{20}{100} \cdot C = A \rightarrow \boxed{B + 0,2C = A}$

$C + \frac{20}{100} \cdot A = 3B \rightarrow \boxed{C + 0,2A = 3B}$

$A + B + C = 68$ (I)

$B + 0,2C = A$ (II)

$0,2A + C = 3B$ (III)

I → II → $(B + 0,2C) + B + C = 68$

$2B + 1,2C = 68$

$B = \frac{68 - 1,2C}{2} \rightarrow \boxed{B = 34 - 0,6C}$

III → $0,2A + C = 3 \cdot (34 - 0,6C)$

$0,2A + C = 102 - 1,8C$

$\boxed{0,2A + 2,8C = 102}$ ← II

$\rightarrow 0,2(B + 0,2C) + 2,8C = 102$

$0,2B + 0,4C + 2,8C = 102$

$0,2B + 2,84C = 102$

$\therefore 0,2(34 - 0,6C) + 2,84C = 102$

$6,8 - 0,12C + 2,8C = 102$

$6,8 + 2,72C = 102$

$2,72C = 102 - 6,8$

$2,72C = 95,2$

$C = \frac{95,2}{2,72} \rightarrow \boxed{C = 35}$

$B = 34 - 0,6C$

$B = 34 - 0,6 \cdot 35$

$B = 34 - 21 \Rightarrow \boxed{B = 13}$

$A + B + C = 68$

$A + 13 + 35 = 68$

$A + 48 = 68 \Rightarrow A = \frac{68}{48} \Rightarrow$

$\Rightarrow \boxed{A = 20}$

Cali tem R$ 20,00

Bia tem R$ 13,00

Coco tem R$ 35,00

Coco − Cali =

35 − 20 = $\boxed{15}$

Cali tem R$ 15,00 a menos que Coco.

$\boxed{\text{Cali tem R\$ 15,00 a menos que Coco. - Letra A.}}$

Exercício 5:

5-) Ailen = 134 | Bento = 115 | Cíntia = 48

↳ x | ↳ y | ↳ z

$$A = \begin{bmatrix} 0 & 3 & 4 \\ 1 & 0 & 5 \\ 2 & 1 & 0 \end{bmatrix} \quad e \quad x = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \begin{matrix} \to 134 \\ \to 115 \\ \to 48 \end{matrix}$$

$$\begin{cases} 3y + 4z = 134 \\ x + 5z = 115 \\ 2x + y = 48 \end{cases} \quad -2 \to \left(\begin{array}{ccc|c} 0 & 3 & 4 & 134 \\ 1 & 0 & 5 & 115 \\ 2 & 1 & 0 & 48 \end{array}\right) \therefore$$

$$\therefore \begin{array}{c} 1 \to \\ \\ -3 \to \end{array} \left(\begin{array}{ccc|c} 0 & 3 & 4 & 134 \\ & & & \vdots \\ 0 & 1 & -10 & -192 \end{array}\right) \sim \left(\begin{array}{ccc|c} & & & \vdots \\ & & & \vdots \\ 0 & 0 & 34 & 680 \end{array}\right) \to \begin{array}{l} 34z = 680 \\ z = \frac{680}{34} \\ \boxed{z = 20} \text{ m} \end{array}$$

$$\left.\begin{array}{l} x + 5z = 115 \\ x + 5 \cdot 20 = 115 \\ x + 100 = 115 \end{array}\right\} \to \begin{array}{l} x = 115 - 100 \\ \boxed{x = 15} \text{ m} \end{array}$$

$$\left.\begin{array}{l} 3y + 4z = 134 \\ 3y + 4 \cdot 20 = 134 \\ 3y + 80 = 134 \\ 3y = 134 - 80 \end{array}\right\} \to \begin{array}{l} 3y = 54 \\ y = \frac{54}{3} \Rightarrow \boxed{y = 18} \text{ m} \end{array}$$

$$\left.\begin{array}{l} x + y + z = ? \\ 15 + 18 + 20 = \boxed{53} \end{array}\right\} \to \boxed{R\$\ 53{,}00} \text{ m}$$

$$\boxed{\begin{array}{l} R\$\ 53{,}00 - \\ \text{Letra A.} \end{array}} \text{ m}$$